

Customer Segmentation using K-Means Clustering

Objective

The objective of this project is to segment customers into different groups based on their purchasing behavior using the Mall Customer Dataset and K-Means Clustering.

Dataset Description

The dataset contains customer information such as:

- Customer ID
- Gender
- Age
- Annual Income (k\$)
- Spending Score (1–100)

Data Preprocessing

- Loaded the dataset using Pandas.
- Checked for missing values and data consistency.
- Selected relevant features:
 - Annual Income (k\$)
 - Spending Score (1–100)

K-Means Clustering

The Elbow Method was used to determine the optimal number of clusters.

Based on the Elbow Curve, 5 clusters were selected and K-Means clustering was applied to segment customers.

Cluster Visualization

A scatter plot was generated using:

- X-axis: Annual Income
- Y-axis: Spending Score

Different colors represent different customer segments, while cluster centroids are highlighted.

Customer Segments

Cluster 1: Low Income – Low Spending

Customers with low annual income and low spending behavior.

Cluster 2: High Income – High Spending

Premium customers who generate significant revenue for the business.

Cluster 3: High Income – Low Spending

Customers with strong purchasing power but relatively low spending activity.

Cluster 4: Low Income – High Spending

Customers who spend actively despite lower income levels.

Cluster 5: Average Customers

Customers with moderate income and average spending patterns.

Business Insights

1. High Income – High Spending customers should be prioritized through loyalty programs and premium services.
2. High Income – Low Spending customers can be targeted with personalized marketing campaigns.
3. Low Income – High Spending customers may respond well to discounts and promotional offers.
4. Customer segmentation enables businesses to improve targeting, customer retention, and marketing efficiency.

Conclusion

K-Means clustering successfully segmented customers into meaningful groups based on income and spending behavior. These insights can help businesses design targeted marketing strategies, improve customer engagement, and maximize profitability.

Screenshots:

